OCP GLOBAL SUMMIT 2026 · TIME APPLIANCES TRACK

Prediction Error Budgets for Time-Scheduled Circuit Fabrics in Distributed AI Training

Circuit switching stopped at per-job granularity. Can we do better?

Sahas Munamala
sahas.munamala@ocproject.net

Ahmad Byagowi
byagowi@turbalance.com

GPUland testbed

25 pre-registered cells · 7×T4 · one clock

WHAT A CIRCUIT CAN ACTUALLY REPAY

Two filters hide the traffic; what survives is circuit-shaped

FILTER 1 · LOCALITY

>75%

of training bytes are tensor-parallel and never leave the NVLink domain. **>99% of GPU pairs exchange nothing.**

FILTER 2 · OVERLAP

1:7

cross-pod oversubscription tolerated in Llama-3: only prefetch-tolerant data-parallel traffic leaves the pod, **hidden behind compute**.

SURVIVES BOTH

14–32%

of fleet GPU-hours are communication with **no overlapping compute**. The collective that defeats both filters — all-to-all — needs full bisection.

Hidden traffic cannot repay a fabric. The case for circuits lives entirely in the exposed remainder.

Three gates: **exposed** · **contended** · **predictable**. The first two are profiling questions and the literature answers them. The third is a *timing* question, and it is the one this talk measures.

Rail-only (Wang et al., MIT+Meta) · MAD-Max (Hsia et al., ISCA'24) · RoCE at Meta Table 1 (Gangidi et al., SIGCOMM'24) · Llama-3 (arXiv:2407.21783)

THE STANDING VERDICT

Circuit switching already won — at per-job granularity

GOOGLE · JUPITER

5×

capacity, −30% capex, −41% power. Reconfigured on TE timescales.

GOOGLE · TPU v4

<5%

of system cost, for 1.2–2.3× from topology shape. One shape per job.

META

0

reconfigurable fabric. Rail-only deletes the spine: −38–77% cost.

SIRIUS

ns slots · dwell/switch ≈ 1

THE GAP · PER ROUND

dwell/switch O(1) · **unmeasured**

PRODUCTION

per job · dwell/switch ≥ 10^6

Jupiter §4.6 on going finer: short-term prediction has never beaten long-term — “Neither has been the case historically.” large-scale ML will change that.

Jupiter Evolving (SIGCOMM'22) · TPU v4 (ISCA'23 §2.7) · Lightwave Fabrics (SIGCOMM'23) · Rail-only (arXiv:2307.12169) · Apollo §3 · Sirius (SIGCOMM'20)

THE MEASUREMENT

Isolating Prediction Error

•**One container per GPU.** NCCL runs over TCP on host veth pairs; tc/HTB classes emulate the switch ports. Two pods (3+4) end each step with an inter-pod all-to-all.

•**Every rank and the controller share one kernel clock.** With no PTP residual, NIC timestamp error, or clock discipline in the loop, synchronization error is zero by construction.

•**Every millisecond of degradation therefore belongs to prediction.** We inject known error ε into t̂ = t_true + ε and measure P(ε) directly across 766 steps and 63 cells.

544 ms	**651 ms**	**2365 ms**	**16.3 s**
pinned-circuit floor	oracle (perfect prediction)	packet baseline (4:1)	all-trickle ceiling

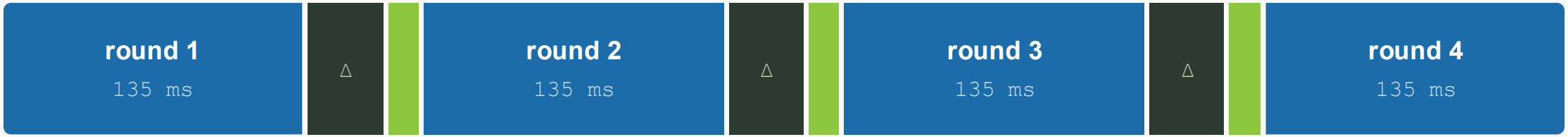


7× NVIDIA T4 GPUs on one EPYC host ("GPUland")

THE MEASUREMENT

Seven GPUs, one clock, and a known error injected into f̂

24 circuits demanded, 6 granted at a time → **R = 4 rounds**, all fired open-loop from one predicted start.



A = 135 transfer + Δ 25 dark + s 10 slack = **170 ms** · dwell/switch ≈ 6 · trickle traffic **40× slower**

Every rank shares one kernel clock.

Sync error is **zero by construction** — so every millisecond of degradation belongs to prediction.

651 ms
oracle

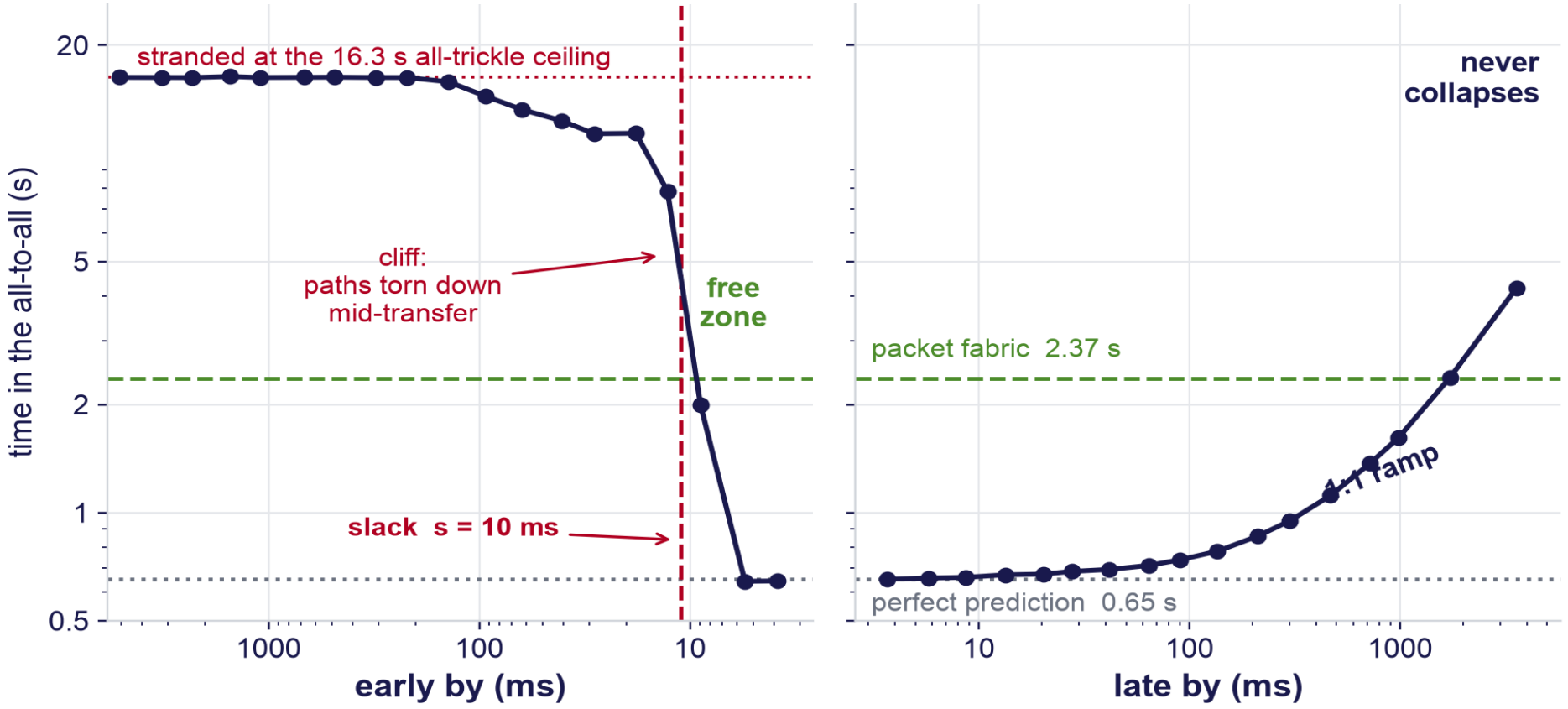
2365 ms
packet

16.3 s
all-trickle

7× NVIDIA T4 on one EPYC host, one container per GPU, NCCL over veths, tc/HTB emulating switch ports · two pods (3+4) · 766 steps / 63 cells, plus 25 pre-registered shared-time cells

THE NUMBERS

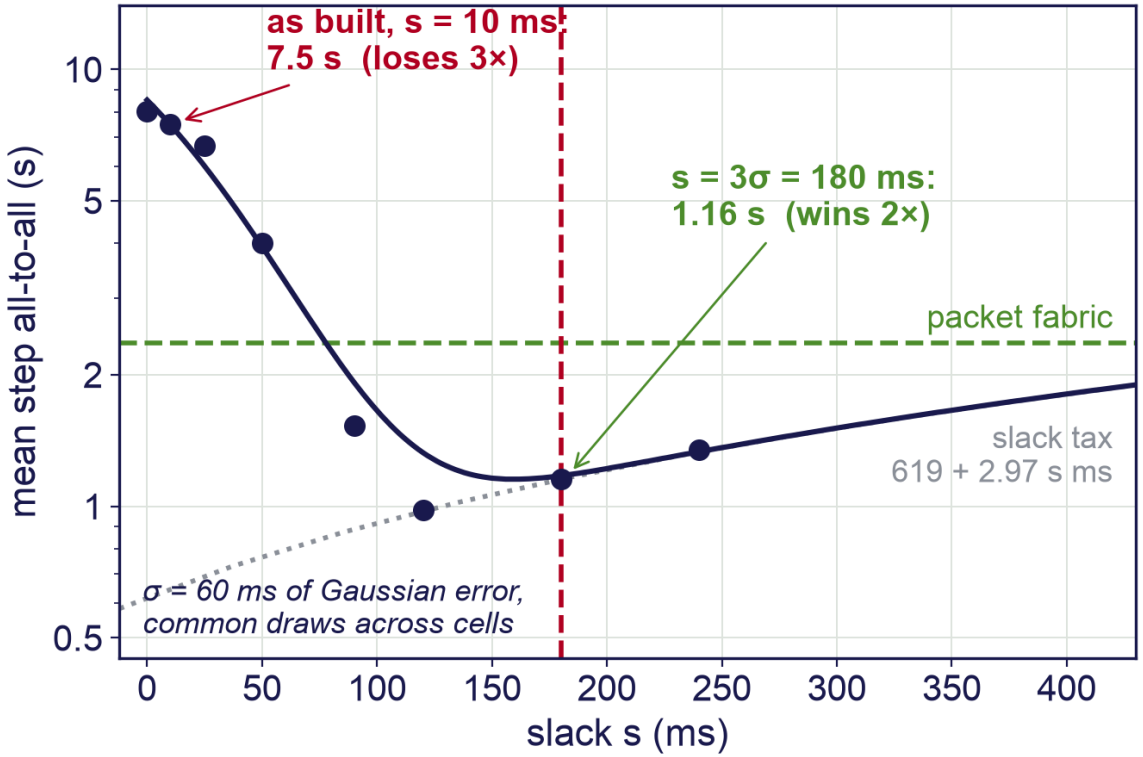
Miss late, pay 1:1; miss early, fall off a cliff



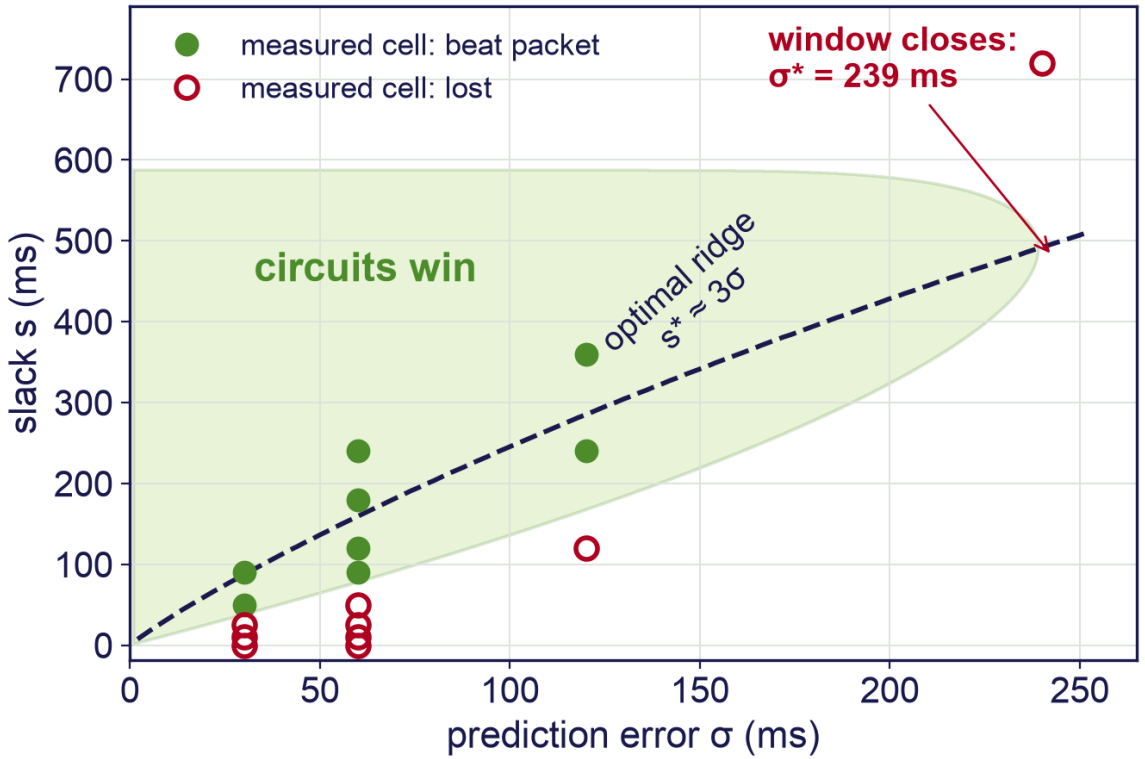
Early tolerance ≈ 11 ms; late tolerance ≈ 1.7 s (~156×). The early cliff tracks slack across an 8× range of round durations, and jitter spends the same budget. One wedged cell excluded cell-wise (fabric artifact, not ε physics).

THE NUMBERS

Size slack to the error



At σ = 60 ms, the as-built 10 ms slack loses 3×; s = 3σ wins 2×, a 6.5× swing.

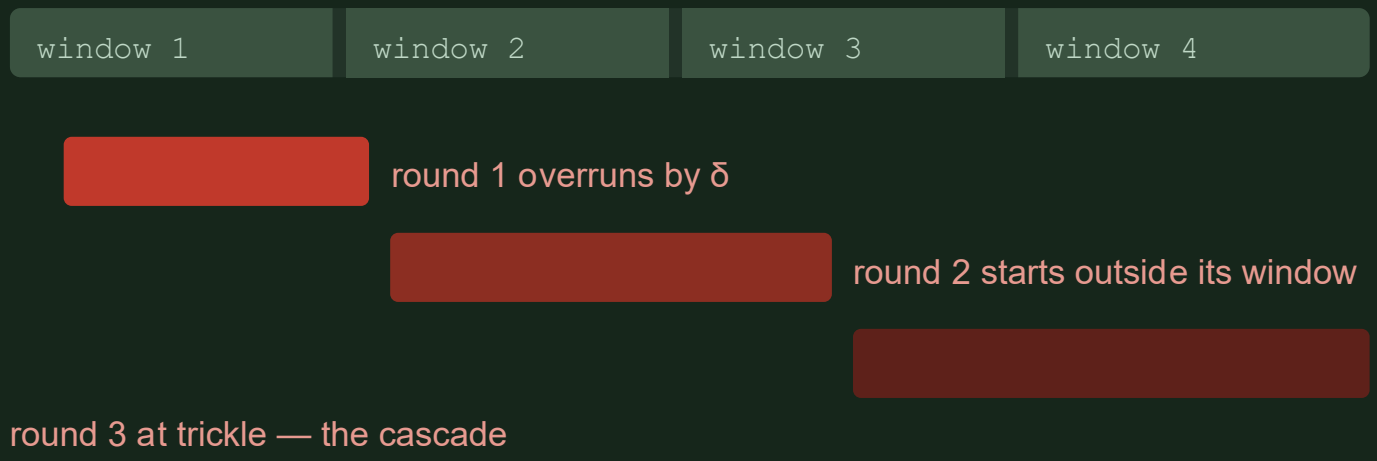


σ moves from 8 ms as built to 239 ms when sized; beyond the close, slack tax alone exceeds packet.*

Is σ ≤ 239 ms achievable? Raw step-time variance at 64K GPUs is already σ ≈ 210 ms (PRISM, arXiv:2510.15596) — the floor for a predictor that merely tracks the mean — inside the sized window. **Live predictor in the loop is the immediate next step.**

THE TURN

The cliff is not the fabric. It is the coupling between rounds.



72%

of collapsed steps cost more than one stranded residual can explain.

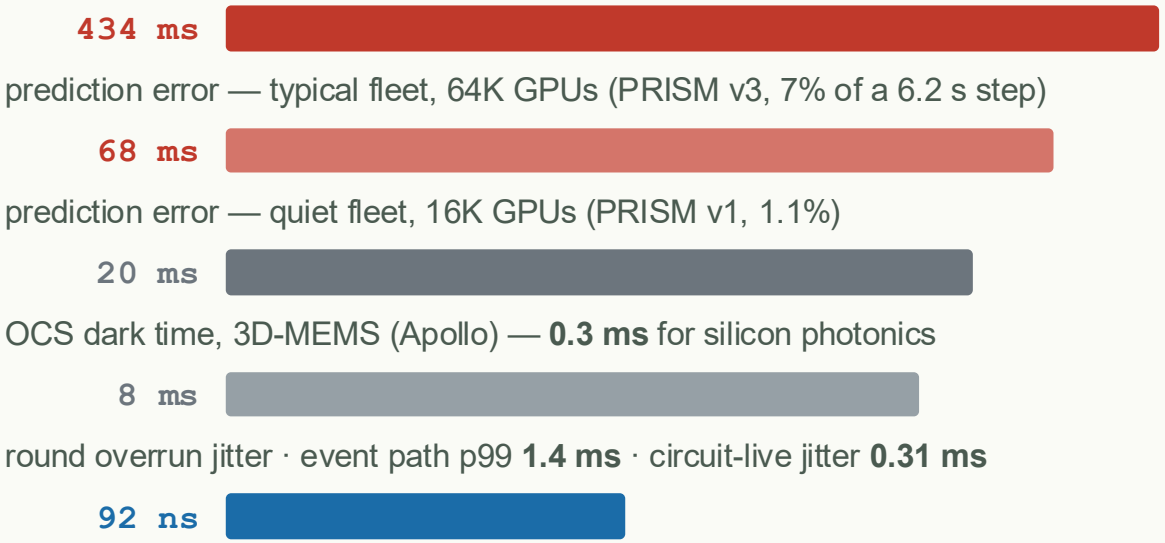
Round k+1 cannot start until round k's wait() returns.

The overrun is cheap. The cascade is the 16 s plateau.

462 collapsed early-error steps across the bias and σ sweeps · an earlier "90%" figure was not reproducible from the CSVs and is withdrawn

THE TIMING BUDGET

Nine orders of magnitude — and sync is five below prediction



For firing circuits, PTP is finished work.

Sync sits five to six orders below the prediction term. It enters the arithmetic in exactly one place — a **tiered** fabric, where $\sigma^2_{eff} = k\,\sigma^2 + (k-1)\,\sigma^2_{sync}$.

The second job is not about accuracy.

Independent round timing does not need a *better* clock. It needs a time base that is **common, addressable in absolute terms, and available to the workload** — not just to the fabric controller.

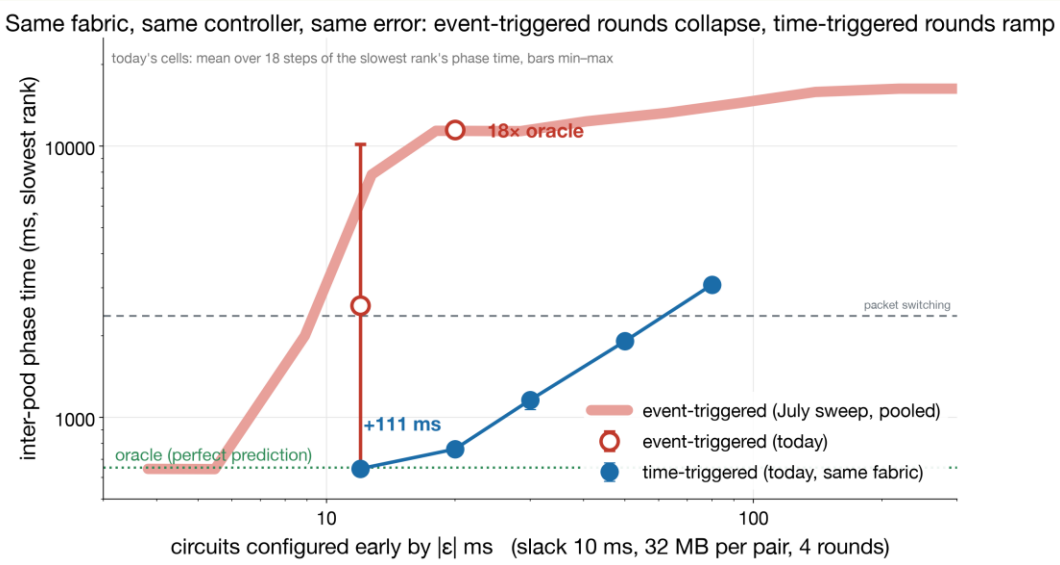
That is an interface ask, not a precision ask.

Collective libraries today schedule on completion events. To schedule on time they need the PHC exposed to the collective, and a round schedule expressed in absolute time.

Bars are log-scaled by eye, not to scale · [L] published, [M] measured on the testbed · results/scale/timing_budget.csv

RESULT 2 · THE MAIN MEASUREMENT

Same fabric, same controller, same error



EVENT-TRIGGERED, ε = −20 ms

11.44 s

on the plateau, every step.

TIME-TRIGGERED, SAME CELL

+37 ms

and free at the oracle: 649 vs 652.

THE LAW

$$39 \cdot (|\varepsilon| - s - 9)^{+}$$

A dead band, then a ramp at the trickle rate. No cliff, anywhere.

25 pre-registered cells, five batches, two seeds, three payloads · slope 38.8 ± 0.1, within 2.8% of the measured trickle ratio · 0 slotted steps on the plateau

WHERE INDEPENDENT ROUNDS HAVE REAL DEMAND

MoE all-to-all is the traffic this was built for

1 · THE EXPOSED CLASS

Meta's own table: **only AllToAll needs full bisection**. Rail-only taxes it 8–11%.

2 · ROUNDS ARE NOT ALIKE

Routing changes every step, so each round carries a **different volume** and finishes at a different time.

3 · THE FIX IS THE SAME

Independent timing makes routing skew **stop being contagious**.

This is an argument, not a measurement.

Every number in this talk is from a **dense** workload. An MoE all-to-all is the next experiment.

RoCE at Meta Table 1 (SIGCOMM'24) · Rail-only forwarding tax (arXiv:2307.12169) · the ByteDance 42.5% straggler figure is from a trace that excluded expert parallelis
do not co-attribute it to MoE

Future Work

- **Measure Open Loop Prediction Performance on Real Rounds of Multiple Different Workloads**
- **Hardware, Real Optical Switches**
- **PTP in the loop**
- **MoE All-To-All**



Thank You!

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AT DEPLOYMENT SCALE · GATED ANALYTICAL MODEL

How much of the circuit win survives the timing error

gain retained, inter-pod all-to-all	16K	64K
tiered OCS, 2 tiers	60%	28%
flat OCS	71%	47%
flat OCS + slots	74%	50%
packet pods + OCS spine	79%	68%
packet pods + OCS spine + slots	82%	70%

Timing error is the budget, not switch speed.

Δ = 20 ms is 5% of the window. Sub-millisecond photonics moves **no row by 2 points**.

The fallback path orders the rows.

A thin packet path caps a stranded round at 7×, below packet — so mixed cannot lose.

Slots: half the guard band.

403 ms down to 168 ms at 16K.

Llama-3 405B DP window, 400 Gb/s, Δ = 20 ms MEMS, typical fleet jitter (σ 214 / 434 ms from PRISM) · model gated on 8 testbed quantities, η the only calibration const
dense, MoE excluded